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Patent Statistics as a Measure of Technological Change

Views on a Doctoral Dissertation

Bjørn Basberg defended his doctoral dissertation *Patenter og teknologisk endring i Norge 1840-1980. En metodediskusjon om patentdata anvendt som teknologi-indikator* (Patents and technological change in Norway 1840-1980. A methodological discussion of patent data used as technology indicators) (Bergen 1984) at the Norwegian School of Economics and Business Administration in Bergen in late 1985. The opponents were Olle Krantz, University of Lund and Hans Christian Johansen, University of Odense.

The dissertation is in six parts: Introduction, Measurement of technological change, Analysis of aggregated data, Industrial sector analyses, Microanalyses and a Summing-up and conclusions. Olle Krantz was allotted the first three parts for comment and Hans Christian Johansen the remainder.

The author gives below a resumé of the dissertation, and this is followed by summaries of the opponent's views.

Bjørn L. Basberg

One method of measuring technological changes is to make use of patent data. In general this may be termed using patents as technology indicators. The object of the dissertation is to evaluate this method of measurement critically by means of analyses of Norwegian patent data from 1840 to 1980.

The background to the work is as follows. Measurement of technological change has engaged the attention of economists, economic historians, historians of technology and research analysts without any uniform method or "yardstick" having been evolved. Numerous methods exist, and most of them have advantages and drawbacks according to the situation in which they are used. In this way patents and statistics of numbers of patents become merely one of many modes of obtaining information about inventions, innovations and technological change in the widest sense. It is easy also to find weaknesses in this method, and little by little a considerable body of foreign literature has accumulated in which the problem is discussed and attempts made to solve it in various ways. However, Norwegian patent material has been subjected to very little previous analysis, and the dissertation accordingly consists in the using of new data to shed light on issues in an ongoing methodological debate.

One problem in trying to decide what sort of information patent data can give concerning technological development is that no other precise measure is available if one wants information above the highly detailed level. There is therefore no yardstick against which to evaluate the patents. The alternative methods are all "indicators",

and it is in the nature of the concept of indicators that they can provide only an incomplete picture of what it is actually desired to illuminate or measure. The mode of evaluation employed in the dissertation is to utilise patent information in *several* different approaches, at varying levels of aggregation, and to compare it with other indicators in the light of the facts of technological and industrial history. One of the main aims is to arrive at a broadly-based evaluation by illuminating the theme from many different angles. By several levels of aggregation, is implied analyses ranging from studies of total numbers of patents applied for and granted over a period of years to patenting associated with industrial sectors, the individual firm or a special technology.

It is a principal hypothesis that patent statistics reflect not only technological but also economic and institutional conditions. A constant theme accordingly is the elucidation of the degree to which this is the case and how patent statistics can be ordered, employed and presented so as to distinguish between these factors. The principal hypothesis is substantiated. For example, in the aggregated analysis of time series for total numbers of patents applied for and granted, it is shown how the two world wars influenced crucially the flow of applications and the numbers of patents granted. During the First World War there certainly seems to have been a real increase of inventive activity, but the peaks that occurred after the two wars were primarily reflections of the institutional barriers that hindered patenting during them. Institutional changes of a different character are changes in patenting law. An example of how the latter may have influenced fluctuations in the aggregated patent statistics is furnished by the great legislative changes of 1885 and 1910, with which are associated pronounced and lasting changes in the proportion of applications granted. A further institutional factor which is important in interpreting the patent figures is the lengthening of the time taken to process applications throughout the period studied. The relationship at the aggregated level between applications and patents subsequently granted also seems to have become weaker over time. Numerous examples are cited to demonstrate that aggregated patents were also influenced by economic factors. It is shown, *inter alia*, that total numbers of applications submitted seem to have fluctuated in step with gross investments after a certain time-lag. It is true that the correlations are of varying strengths during different part-periods, but the results here point in the same direction as those in, for example, J. Schmookler's studies.¹ Aggregated figures for Norwegian patenting in the United States show links not only with economic variables such as exports and cyclical trade movements, but also with an institutional factor as the characteristics of the American patent system.

Also in the industrial sector analysis it is found that in several ways both sector variations and developments in patenting over time reflect in part both economic and institutional factors. However, patent statistics can be ordered in different ways so as to correct for the influence of such factors. Attention is focused primarily on the relationship between patenting and employment in the various sectors. Employment figures have been used as an indicator of economic significance, and wide variations between the sectors and substantial changes over time are found in these compari-

¹ Schmookler, J., *Invention and Economic Growth*, Cambridge, Mass. 1966, p. 14.

sons. No proportionality is found between the size of a sector measured in patents and its size measured in terms of employment. In this respect the results arrived at are not in agreement with those of for example Schmookler. On the other hand it is considered that variations in a patent/employment ratio can reflect differences of inventive activity between the sectors. Attempts have been made to explain these variations by introducing both economic and other technological variables. In particular there appears to be a strong correlation with expenditures on research and development (R & D). This is a relationship which is demonstrated in a number of studies and which thus appears to be valid for Norwegian conditions as well. But the relationships are not unambiguous, and the sector differences revealed in the relationship between R & D and patenting are presumed to reflect differences in patenting practices and patent policy in the sectors. While the chemical industry, for example, has emerged in the postwar period as the paramount patent sector, its position is less dominant when measured against the relative share of total R & D expenditures. This may indicate that this sector uses the patent system more actively than the real level of inventive activity would call for.

The microanalyses have been designed somewhat differently from the more aggregated analyses. Here too, however, it is shown that the use of patent statistics as technology indicators is problematical. For example, the conclusions drawn on the basis of analyses of patents associated with the so-called Söderberg electrode and the slipway on whale factory ships are striking in terms of the picture they give of the innovation process. While the number of patents granted to the firm which produced the electrode seems to have reflected the diffusion of this technology over time, the peak year for the number of patents related to the slipway reflects the point in time of first innovation.

What primarily emerges from the discussion is that very many factors have to be introduced in order to explain patenting. For the purpose of singling out a type of patent information which unequivocally reflects inventive activity or technological change, it is necessary to go through a "stripping-down process" in which the effects of institutional factors, patent policy, patenting practices and economic factors are all eliminated. This process demands such a thorough and detailed knowledge of the patent system, and of condition and economic development of the industry, that it may be difficult to make the patent statistics usable as technology indicators.

The picture thus drawn is relatively gloomy with respect to the chances of using patent statistics as technology indicators. But at the same time a number of factors are identified which make patents an interesting source of information. Numerous examples are cited of links between the trend of patent statistics and technological change. The microanalyses especially seem to show that patent data can constitute valuable information at this level, for example by illustrating how inventive activity moves from sector to sector according to where the bottlenecks or technical production problems present themselves. The studies seem to show that the data are less liable to distortion when we shift the analysis to the micro level, which is not of course a feature unique to patent data. It is also shown that patenting is linked to the industrial *milieu*. There has been a trend away from patenting directed by individuals and towards patenting directed by firms, but the link with industrial firms has always been strong. Patenting is also diffused geographically according to the location of in-

dustry rather than that of population as such. It is further demonstrated that there are connections between patent data and other technology indicators such as capital intensity, productivity and R & D. The discussion of these connections, especially those with R & D, discloses an opportunity for correcting erroneous interpretations of the patent statistics. The manner in which the patent data are deployed, however, will be important for the result and the interpretations. Weighing will be more important than counting, and this can be effected by using patent information in combination with more traditional historical analysis or with other indicators as a partial indicator. Finally, the patent data appear to be capable in some degree of reflecting the technological aspects of leading economic sectors. However, which types of patent data are selected or utilised is absolutely crucial. It is not enough to study patents in Norway from Norwegian sources in isolation. Patenting in Norway by foreigners and Norwegian patenting abroad both provide valuable information. In particular it would appear that Norwegian patenting in the United States represents data of high quality and of high informational value.

Olle Krantz

According to the title, the dissertation deals with patents and technological change in Norway 1840-1980, and the subtitle amplifies this: there is to be a methodological discussion of patent data as technology indicators. The main theme is formulated many times in the book: "We are asking whether patent statistics are in any way unique as a source of information in the study of technological change, and we are asking what can be involved in utilising this type of information in a 'cautious manner'" (p. 79). When a dissertation is so orientated, it is important to discuss the validity of the indicator, i.e. whether it measures what it is intended to measure, or in this case indicates what it is intended to indicate.

An economic-historical validity test can be designed in various ways. Generally speaking, however, it is an advantage if there is a clear statement at the outset of what is to be studied, in what theoretical context it is to be viewed, and how the empirical measures are to be constructed, i.e. how they are to be adapted to suit the formulation of the problem. In this way criteria of validity will also have been obtained.

If a validity test is to be made of measures which are not constructed as stated but which nevertheless can be presumed to be usable, the character of these data must be made clear. The measures ought to be congruent with the theoretical frame of reference, implying *inter alia* the same level of aggregation. One feature of the validity test can also be to check the chronological sequence of relevant variables in relation to one another.

Basberg's attitude to this complex of problems is not wholly clear from the dissertation. Validity is mentioned explicitly at a number of points, but it is not made entirely plain how validity is to be assessed despite the fact that basically the entire work constitutes a treatment of the validity problem.

Indicators and measures

An important feature of the dissertation from the validity standpoint is the discussion of the concepts of "indicator" and "measure" (p. 30ff), where it is argued that most methods can measure technological change only indirectly. For the most part, therefore, all that can be done is to make use of some indicator, which implies that it is only possible to obtain a rough idea of developments and tendencies. But it is seldom or never possible to obtain an exact measure of technology. An indicator, says Basberg, is imprecise, and an indicator of technology is associated with one of the stages of the innovation process and not with the process as a whole. If an indicator is used the level of ambition is lower and the danger of false conclusions less than if a precise measure were being used.

The question is, then, what is to be regarded as a precise measure? Possibly it might be what is called a direct measure (p. 31). It could then be interpreted - to cite an example - so that a precise measure of the production of a certain type of iron ore is the quantity mined during a certain period. But such measures do not enable us to reach any level of aggregation. Probably no one would protest, however, if the index of industrial production were termed a direct and perhaps precise measure of industrial production. In Basberg's terminology it would, however, have to be called an indicator.

This is a key issue in the discussion of validity. If an "objectively correct", always valid measure of an aggregated nature exists, it is a precise measure. But do such measures exist? Take for example the volume of the national product or the volume of industrial production. All the quantities comprised therein may be measured accurately and exact information about all prices selected may be available. But these data can be aggregated in a number of different ways, usually with different volume series as a result. Which is then the objectively correct and which is consequently the true, exact or precise economic measure? From these starting points all measures have high reliability. But what about validity? The latter cannot be assessed in the specific instance unless it is clear what it is to be assessed against. This must be clear from the formulation of the problem and the theoretical considerations related thereto.

The argument likewise applies at appropriate points to patent data as a measure of technology in studies of economic history. In practice what are involved here are economic measures, though generally not weighted as in the case of the volume of industrial production. But there must be certain explicit criteria or models which can form a basis for an assessment of validity. If this is so, the difference between the measure and indicator concepts must be minimal, and it could even be argued that these terms are synonymous.

Basberg's definitions

What is to be measured with respect to technological change is innovative activity, and as has been remarked, by indicator Basberg means a measure which can be linked to some definite element of the innovational process. A starting point for assessing the indicator's relevance must be innovation models containing such elements, and a number of models are specified in the dissertation. They are elaborated

to varying degrees and contain "stages" such as research, development, production and diffusion, usually with a number of activities as subgroups at every stage. However, the models belong to the micro level. The activities are of the kind in which a company or firm engages. This micro orientation on the theoretical plane may have contributed to mistrustfulness on the part of the author towards using patent data on the macro level, and it may have caused studies of aggregated figures to have been less intensive than might otherwise have been the case.

It is also possible that this micro orientation at the starting points may have given rise to a certain lack of clarity with regard to the concept of innovations as against that of inventions, and the role and import of the patents in respective instances. Perhaps a certain persistence in sticking to the level of the firm is also responsible for the fact that inventors and innovators outside the factory gates have been given too small space. (This means individual patents as opposed to company patents.)

A heavier emphasis on the macro aspects in the theoretical discussion could have helped to make the concepts somewhat clearer, e.g. by drawing a boundary between invention and innovation. Inventions can be regarded as technically and/or scientifically interesting to a considerably higher degree than innovations, which are economically interesting. In economic history, attention may be paid to the latter when patents are being regarded as indicators or measures of technological change.

One perhaps gets some idea of Basberg's view of this distinction from the way in which in various situations he expresses views such as that patents led to innovations or were the basis for innovations (e.g. p. 54 and 56). This may suggest that he thinks that patents are taken out on inventions which are technically interesting. But he also says (p. 53) that one must also reckon that the inventor, or whoever is backing him, sees economic potential in the invention.

Schmookler states: "We shall require that an invention be not only new but also prospectively useful".¹ His discussion then amounts to showing that technical change is endogenous in the economic system and that patent data are usable as a measure of inventions thus defined. Consequently he invests the concept of invention with a meaning different from that just given. Schmookler's concept of invention, then, comes quite close to the concept of innovation in the sense indicated above, while the use of the concept in Basberg's book is not entirely clear.

What do patents show?

Basberg propounds an argument (p. 54ff) which betrays a certain degree of scepticism towards patent data as a measure of innovation. The question is posed whether patents describe or measure basic inventions, improvements or less important inventions, or diffusion. But he retains his micro perspective, and one of the points he argues is that the earlier in the development of a product (i.e., the nearer the idea stage) the object is when the patent is taken out, the smaller the probability that the patent will be put to use. But is there any particular likelihood that the patent will be taken out before it is clear to the patentee that the patent is potentially profitable?

¹ Schmookler, J., *op. cit.*

If there are considerable variation in the commercialisation of the patent, it would indicate variations also in the moment of patenting during the creative process. However, Schmookler finds the degree of this commercialisation to be surprisingly high and even, and Basberg too cites data indicating this. (p. 53) It is scarcely remarkable that there are qualitative differences between different patents, but it is not wholly clear from Basberg's analysis whether they are so wide or of such a character as to make patent data unsuitable as a measure of technological change at the macro level.

So Basberg asks himself: When Schmookler analysed correlations for peaks and troughs in the deviations from the trend for patent and economic variables, what was he really analysing? Basberg introduces basic inventions into the discussion, justifying this on the ground that these are the truly significant ones although they may be preceded and followed by multitudes of patents taken out on components or other items of lesser importance. A possible scenario, Basberg believes, could be that the patent series and the series for basic inventions have simultaneous peaks and troughs. But where then to place the innovations which are central to the scenario? Schmookler is said not to have distinguished between patents which remained inventions and those which became the basis for innovations. Basberg believes that if this distinction is made, it would be possible for basic innovations to vary directly counter to the patent figures, i.e. for the two variables to follow inverted paths.

With regard to basic inventions, Basberg cites an article by van Duijn² in which a total of 80 basic inventions appearing since the first half of the nineteenth century are specified. These are divided into 13 segments of the economy, which are termed "growth sectors". These 80 inventions are interesting from the standpoint of technological history, but how do they look economically in the short term? In the longer term their significance is indisputable, of course. It is doubtful whether these basic innovations describe technical development as Basberg claims. According to all innovation models, a great deal takes place after the innovation stage. It has even been asserted that it is the small steps that constitute the interesting technical development.³

Another interpretation could be that in the initial stages basic innovations do not have many advantages, if any at all, because they are technically incomplete in relation to what they are to replace or what is being demanded. But they have a large potential for economic and technical improvement, and the prospects of profit cause development work to take place on a broad front, which gradually results in greater economic benefits - and more patents. This is an innovative activity which must be regarded as economically interesting. From this starting point, the trend of patents in total could result in peaks that coincide with innovation performance in the manner

² van Duijn, J.J., *Fluctuations in Innovations over Time*, *Futures* 1981:4. A more detailed discussion is found in van Duijn, J.J., *The Long Waves in Economic Life*, London 1983.

³ See e.g. Rosenberg, N./Frischtak, C.R., *Technological Innovation and Long Waves*, *Cambridge Journal of Economics* 1984.

indicated by Schmookler. Thereby the economically most important technological change would also be described, but perhaps not the technically most interesting.⁴

The author's conclusion of the discussion runs as follows: "The use of patent statistics in an industry, in a period of time, or in a country, therefore cannot be taken for granted as an indicator of what one is trying to study. All studies using aggregated figures should endeavour to clarify what it is that the figures express in the particular industry or particular period of time being studied." (p. 57)

It is easy to agree with this; one has to be cautious with one's data and conclusions. But one should therefore work intensively and critically - theoretically and empirically - with variables. Basberg's work, primarily with micro models, leads him to be deeply sceptical of patent data as technology indicators on the macro plane, but analysis based on theoretical considerations at this level is generally lacking.

Aggregated patent data - what they may or may not tell us

Basberg's dissertation contains a presentation and analysis of Norwegian aggregated patent data for the period 1840-1980. After having given a number of different series, he tests (p. 89) whether these give a picture of inventive and innovative activity in Norwegian manufacturing industry, i.e. of its technical progress. He thus limits the discussion by confining it to industry despite the fact that many patents relate to other sectors of the economy.

Then the question is raised whether any one patent series, e.g. patents applied for or granted, is better than the others when they are to be compared with series for economic change. After a long discussion he concludes (p. 104) that several types of data are usable. But he believes that total numbers of applications submitted are least liable to fluctuations of e.g. an institutional kind. Therefore they should provide, *inter alia*, the best description of technological change in Norway.

In general terms a careful analysis ought to be made in order to ascertain whether any kind of regular pattern can be discerned chronologically and in relation to other economic variables. Furthermore, attention ought to be paid to several series both for patents and the economy. Some calculations of correlations between a couple of variables are hardly sufficient. Patent applications by Norwegians, for example, may follow a different temporal pattern with regard to fluctuations of different length from that displayed by foreigners' applications. The series can be influenced by or have connections with economic variables in different ways, and as well as this the connections may change between different periods.

Basberg tests possible connections between total numbers of patent applications and one economic variable, viz. gross investment. These variables have rising trends; a calculation also shows a strong correlation between the series when the trends are not removed, 0.82 without lag and 0.86 when the patent series is lagged 3-4 years, which, of course, is not a significant difference. But there is no correlation at all when the trends are eliminated, which in this case implies that the correlation is calculated between series for annual changes.

⁴ It is possible that the correlation fluctuates over time in a systematic way, a contingency which, however, is not discussed by Schmookler.

Calculations are also made for subperiods: 1865-1920, 1921-1939 and 1946-1980, i.e. the "classical" division into prewar, interwar and postwar periods. The interwar period stands out as divergent from the others. For the original series the correlation is much weaker than for the other periods and the significance level is low. On the other hand the correlation for the detrended series is stronger, and in contrast to other periods the patent series during the interwar period are said not to follow in the wake of investments.

It should be possible to analyse the aggregated figures in somewhat greater detail. Firstly, we may ask whether the division into periods is justified in terms of economic history. The different pattern which Basberg finds for the interwar period compared with other periods perhaps began earlier or perhaps it ceased to be valid sometime during the period 1920-1939. A scrutiny of the series suggests that a more appropriate division might be the period up to the 1880s, the 1880s to about 1900/1910, 1900/1910 to about 1930, and from the latter date to the end of the 1940s. Secondly, there can be different relations with the economic variables for different patent series. Foreign applications in Norway have for instance a different pattern of change from Norwegian applications. Thirdly, investments are not necessarily the only appropriate economic variable for all such comparisons. Production variables may also be important. Basberg is correct in arguing that Schmookler uses investment variables. But in highly aggregated patent variables there are not only patents relating to production methods but also patents that relate to products. In other words, both method and product innovations are involved.

Patent activity thus seems to be linked to the economy, and this is also Basberg's conclusion. But different economic forces further bear upon different aspects of development during different periods, and the world wars might not act as boundary markers for this division into periods to any high degree.

Norwegian patenting in the United States

The source of inspiration for this part of Basberg's work seems to be the studies of Pavitt and Soete⁵, in which it is claimed that patenting in the United States can furnish valuable information on technical change in the country of origin. Pavitt and Soete assume that all countries have the same propensity to patent in the United States in relation to the magnitude of their total innovative activities. The assumption is justified on the ground that the United States offers such a large and sophisticated market that all countries want to export to it, which also provides an incentive to take out patents there.

Pavitt and Soete support their contention that patenting in the United States reflects technical change in the country of origin⁶ with cross-sectional data for various countries for 1975. There is a high correlation between the ratio of United States patents to all patents and the proportion of costs for R & D in a number of

⁵ Pavitt, K./Soete, L., *Innovative Activities and Export Shares: Some Comparisons between Industries and Countries*, in Pavitt, K. (ed), *Technical Innovation and British Economic Performance*, London 1980.

⁶ *Op.cit.*, p. 40.

sectors. (No Norwegian data are included, however.) Moreover they find connections - also based upon cross-sectional data - between exports to the United States and patents in that country for a number of industrial branches in various countries. Finally, there is a very short section on patenting in the United States by six important countries since 1883. Broadly speaking, all that this discloses is that a comparison between the countries shows a relative decline for England.

The question, then, is whether this analysis justifies an historical study focussing upon Norwegian patenting in the United States. The criterion was the large and sophisticated United States market, but this has not always been equally large and equally sophisticated. As Basberg points out in his book (p. 21, see also p. 64f), this situation has fluctuated in that other countries previously held the position enjoyed by the United States today with regard to both the size and the degree of sophistication of the market. Norwegian patent data show that Great Britain was a particularly large market in earlier times and that Germany too was important, as was Sweden likewise. In addition, Great Britain at least, and possibly also Germany, can probably be regarded as sophisticated if this implies technical prominence. A study of patents granted in Norway reveals a similar picture of the two countries' importance, though with Germany becoming more and more prominent. Only during the period since the Second World War has the United States asserted its dominance in this respect. According to Basberg, however, it is difficult to find data from earlier years for Norwegian patents in other countries - which does not, however, necessarily mean that it is impossible; use could be made of these countries' own patent statistics.

Patenting in other countries is probably well worth analysing in order to shed light in technological progress in a particular country such as Norway. But it would have been better to try to analyse Norway's patenting activity in different countries during different periods according to how lively it has been. If at the same time these countries' own activities in Norway had been brought into the analysis, it is possible that important conclusions might have been drawn concerning technological change in Norway in relation to international technological interchange.

Concluding remarks

My views on the macro-orientated parts of Basberg's study of patents can be summed up by saying that from the economic history standpoint one might have expected more intensive work on the material. It is hardly possible on the basis of this study to conclude that patent applications at a highly aggregated level can be regarded as uninteresting in the context of economic history.

One may ask oneself, however, how much can really be contributed to research by new macro studies or studies in general based on patent data. Perhaps Rolf Adamson is right when he says thus: "The total judgment ought to be justified that investigations of patent statistics have as yet more confirmed what was known before

about how the whole economy works than given any distinct additions to our knowledge of a special character."⁷

Hans Chr. Johansen

In the many theoretical works on economic growth which have formed a feature of economic theory in the last thirty or forty years, the pattern of growth has been explained firstly in terms of the increase of the workforce and the quantity of real capital, and secondly as a residual, which in most empirical works has revealed itself as the most important single factor. This residual, often called total factor productivity, has been interpreted *inter alia* as a manifestation of a better educated workforce, as more productive real capital per unit of money invested, the effect of inter-sectoral shifts in the pattern of production, and a better organisation of the production process. Others, however, have been critical of the model's specifications and the data used in the analyses. In general it is true that there is considerable disagreement over what should be considered as ingredients of the residual, which has been termed, ironically, "a measure of our ignorance".

There can scarcely be any doubt, however, that the rising productivity of real capital consequent upon the use of more advanced technology must be one of the factors explaining the course taken by economic growth over the last hundred years, and it is therefore natural that there has been much interest in devising more direct measures of the significance of technological progress than merely regarding it as an unknown ingredient of a residual.

Nevertheless, once one moves away from the micro plane it has been very difficult in practice to produce measurable quantities. It may well be possible to work out how much more effective one machine is than another in producing a certain article, viewed in relation to the prices of the machines, their use of labour and other factors; but at a more aggregated level considerable problems of measurement arise, engendered by the different ages, uses and sizes of the machines, and also by associated difficulties in finding suitable deflators.

As time has gone on, therefore, efforts have been made to find other and indirect measures of technological progress through the use of already-existing time series. The idea has been conceived that patent statistics might be capable of constituting such a "proxy". In most countries such statistics exist in published form for the last hundred years or so, and most people will no doubt immediately link patenting with technological innovation.

It is, however, evident from the considerable international literature on the subject during the past twenty years that patent statistics on the macro-level are not a problem-free indicator of technological progress, and this is confirmed by the study which Bjørn Basberg has made of Norwegian patent statistics.

⁷ Adamson, R., *Glimpses from the History of Industrialization in Sweden, NIR Nordiska Immateriellt Rättsskydd*, 1985, p. 548.

Among other features of the international discussion have been analyses of technological progress measured by patenting in relation to other economic factors. Are technological changes to be regarded as exogenous, emerging independently of economic conditions in the society, independent variables which have influenced other factors after a certain "economic lag", or shall we regard them instead as endogenous factors? If bottlenecks occur in the production process, because of changes in the pattern of demand for example, will this lead to increased efforts to evolve a new technology, and after some time has passed - a "technical lag" - will these efforts perhaps bear fruit? This scenario too has been tested in the Norwegian study, and a slight tendency towards a technical lag of about ten years has been demonstrated.

A considerable part of Bjørn Basberg's work on the Norwegian patent statistics, however, has been conducted on a more disaggregated plane, and this may afford grounds for a number of remarks of a methodological character.

One aspect of the disaggregation is of a division into industry groups, the idea of which is firstly to discover which industries have used patenting most intensively and secondly to investigate whether there is a link between the extent of an industry group's patenting and other measures of rapid technological progress and aggregate growth.

In this connection one may begin by asking how to translate the classification criteria of the patent statistics into an industrial classification. The main criterion for this problem may be to work out which industry derives the most benefit from the invention in question, but this is far from always unambiguous.

Let us take an example:

NN, an engineer employed by a *firm of engineering consultants*, takes out a patent on a type of cement kiln of higher productive capacity than previous ones and sells the licensing rights to an *engineering works*, which manufactures the kiln, which is put into service at a *cement works*.

Three occupational groups are involved here, viz. a service firm, an establishment in the iron- and metalworking industry, and a firm manufacturing non-metallic mineral products. In which of these three categories does this technological advance take place in practice? Presumably primarily in the cement industry, and for this reason almost all authors who have worked on patent statistics usually assign the patents to the user industry. But this has the drawback that all patents used for producing goods which enter directly into consumption can not be included in the analyses. Moreover, there are some machines which cannot be unambiguously assigned to an industry group, e.g. engines, and they may very well be thought to have an appreciable influence on productivity rises in an industry. Finally, problems arise from the fact that both goods and production methods can be patented. This last objection is particularly relevant to the pharmaceutical industry - one whose patenting is very important - in which practically all patents relate to production methods. Since, however, the patent statistics do not show which ones are concerned with production methods, it must be concluded in these instances that the patent has to do with the production of a consumer article and therefore does not represent a technological advance in the pharmaceutical industry. Nevertheless one often sees them being assigned to the chemical industry despite the "user principle".

Light can also be shed on this problem with the aid of the example above. If this patent had instead shown a more rational way in which a cement kiln could be produced with the same productive capacity as previously employed, then the productivity effect would have accrued to the engineering works. But when, as happens in these types of analysis, large groups of patents are assigned *en bloc* to an industry, regard cannot be paid to such individual cases.

The result is therefore that in most studies probably too few patents are assigned to the engineering industry and too many to the electrical industry and parts of the chemical industry.

The patent classification thus revised is used in the Norwegian study for a careful scrutiny of the volume of patenting in the individual industrial sectors and its significance over time. Not unexpectedly, the engineering, electrotechnical and chemical industry groups have a patenting intensity throughout the entire period from the end of the last century until the present day which is somewhat higher than their significance in Norwegian industry, but the importance of several of the leading sectors does nevertheless change with time. Thus, in 1890 it was the graphic industry which was relatively the most important holder of patents, and the textile and clothing industries were likewise prominent. Substantial structural realignments, however, took place during the interwar years. It was now the electrotechnical industry which was the dominant patentee, being succeeded as leader by the chemical industry in the 1970s.

Alongside this more descriptive scrutiny, moreover, an analysis is carried out of the patenting intensity of the sectors compared with their production levels, their outlays on research and development, the average size of firm, share in exports and real capital per employee of the respective sectors. The object of this analysis is, firstly, to discover whether there is any common variance between these quantities, which can all be interpreted as "proxies" for the technological level. This part of the study shows in most cases relatively low correlation coefficients, which can be attributed partly to the above-cited problems in assigning the patents to the sectors to which their technological significance accrues and partly to the fact that the factors mentioned are all somewhat uncertain measures of technological progress.

The analysis is also used for testing a possible causal relation, e.g. between the volume of patenting and R & D expenditure per employee. This is somewhat problematical on the basis of the data employed. If one takes the above-cited example of the cement kiln as a starting point, it may be expected that the research costs will be defrayed by the engineering firm and development costs possibly by all three sectors, and that according to the mode of classification selected the patent will be assigned to the cement works. A correlation analysis based on cross-sectional data cannot therefore give a meaningful result in such a case.

Even if this example, with its distinction between the research and the user sector, should be the exception, there are nevertheless other problems with the interpretation of possible causal relations. For example, when one finds a certain positive correlation between a sector's average size of firm and its volume of patenting and is trying to view it as something more than two measures of the technological level, which is cause and which is effect? It is no use attempting to lag the one variable in relation to the other, for the internal shifts between the sectors

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take place so slowly that the correlation coefficients with lags of +2, +1, 0, -1 and -2 years, for example, are of about the same order of magnitude. A logical preference for one of the variables as the dependent one may be hard to justify. It could be contended that large firms invest more in R & D than small and therefore achieve more results, which are patented. But it could equally be claimed that industries which avail themselves of patents developed in other establishments thereby achieve a monopolistic position and become large - which reverses the causal relation.

As a final methodological factor which is probably particularly relevant when using the patent statistics of a small country, we may cite problems concerning the degree of certainty with which "nationality" can be established in the patent statistics.

If we want to compare figures for Norwegian patenting with other Norwegian technology indicators, we must necessarily study the Norwegian patents granted to Norwegian firms, but if we follow the practice of the official statistics by using the patentee's address in Norway as the criterion of the firm's being a Norwegian one, then numerous cases will be encountered where it is obvious that the patent is actually a foreign one. This is most palpable in the cases of multinational companies applying for patents through a sales or service department in Norway. Thus, one of the major patentees in Norway in the 1920s was Western Electric, with a long succession of patents connected with telephonic technology. These are reckoned in the Norwegian statistics as patents of Norwegian origin, because the American company had a small subsidiary company in Norway which submitted the patent applications. Indeed, going purely by the patent statistics one would have to conclude that the leading countries in the field of telephone technology were Norway and Belgium. The latter country housed the European headquarters of Bell Telephones in Antwerp, and it was from there that the parent company's inventions were patented in European countries.

Thus, as will be seen from the remarks above, and as will also have appeared from many of the results of the Norwegian study, patent statistics are not really a reliable short cut for finding a technology indicator. On the other hand, studies of particular patents or groups of patents within a limited field and of their economic significance for the expansion of a firm or branch of industry can probably yield much more solid results. Bjørn Basberg has shown two such examples in his dissertation, viz. the development by the Norwegian ELKEM company of continuous self-burning electrodes for electrical blast-furnaces, and patents relating to slipways for whale factory ships. These are two good examples of significant Norwegian inventive activity with important economic consequences for the industries concerned.